Chapter 02 Part 1: Chemical Bonding

1.

Element	Electronegativity
H	2.1
C	2.5
S	2.5
F	4.0
Cl	3.0
Si	1.8

On the basis of the information above, which of the following arranges the binary compounds in order of increasing bond polarity?

(A) $CH_4 < SiCl_4 < SF_4$

(B) $CH_4 < SF_4 < SiCl_4$

(C) $SF_4 < CH_4 < SiCl_4$

(D) $SiCl_4 < SF_4 < CH_4$

2. Which of the following has the bonds arranged in order of decreasing polarity?

(A) H−F > N−F > F−F

(B) H−I > H−Br > H−F

(C) O−N > O−S > O−Te

(D) Sb−I > Sb−Te > Sb−Cl

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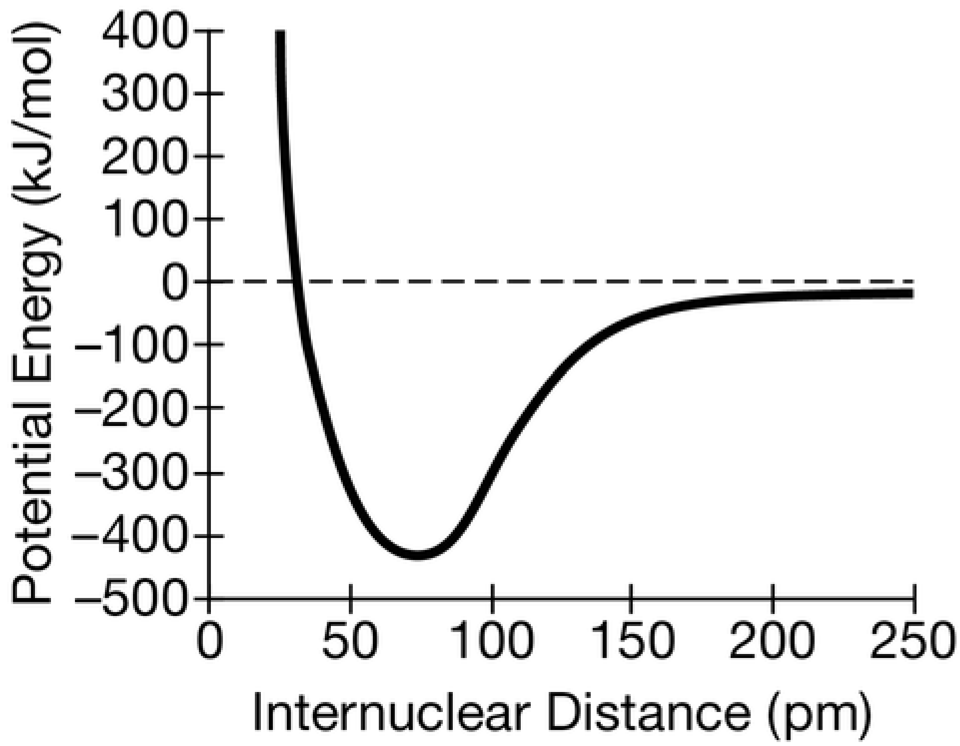
3.



Which of the following can be inferred from the diagram above that shows the dependence of potential energy on the internuclear distance between two atoms?

(A) The atoms form a bond with a bond length of 25 pm.

(B) The atoms form a bond with a bond length of 75 pm.

(C) The net force between the atoms is attractive at 25 pm.

(D) The net force between the atoms is attractive at 75 pm.

4. Which of the following scientific claims about the bond in the molecular compound HF is most likely to be true?

(A) There is a partial negative charge on the H atom.

(B) Electrons are shared equally between the H and F atoms.

(C) The bond is extremely weak.

(D) The bond is highly polar.

5. Which of the following compounds contains both ionic and covalent bonds?

(A) SO_3

(B) C_2H_5OH

(C) MgF_2

(D) H_2S

(E) NH_4Cl

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6. Two pure elements react to form a compound. One element is an alkali metal, X, and the other element is a halogen, Z. Which of the following is the most valid scientific claim that can be made about the compound?

(A) It has the formula XZ_2.

(B) It does not dissolve in water.

(C) It contains ionic bonds.

(D) It contains covalent bonds.

7. The elements C and Se have the same electronegativity value, 2.55. Which of the following claims about the compound that forms from C and Se is most likely to be true?

(A) The carbon-to-selenium bond is unstable.

(B) The carbon-to-selenium bond is nonpolar covalent.

(C) The compound has the empirical formula CSe.

(D) A molecule of the compound will have a partial negative charge on the carbon atom.

8. Which of the following graphs correctly shows the relationship between potential energy and internuclear separation for two hydrogen atoms?

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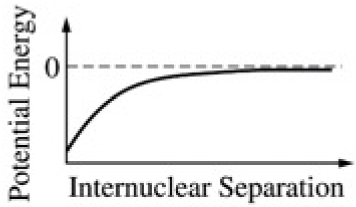
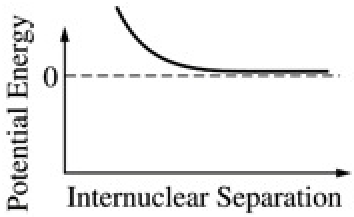
(A)



(B)



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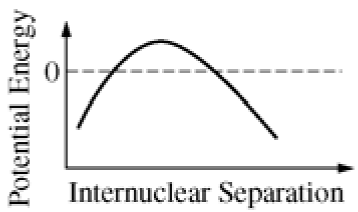

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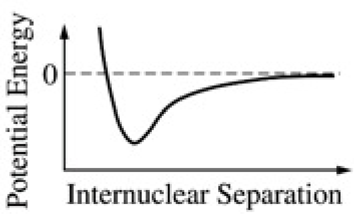
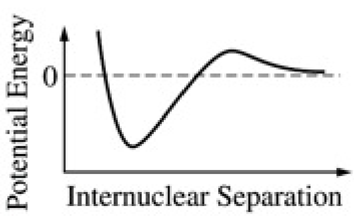


(E)



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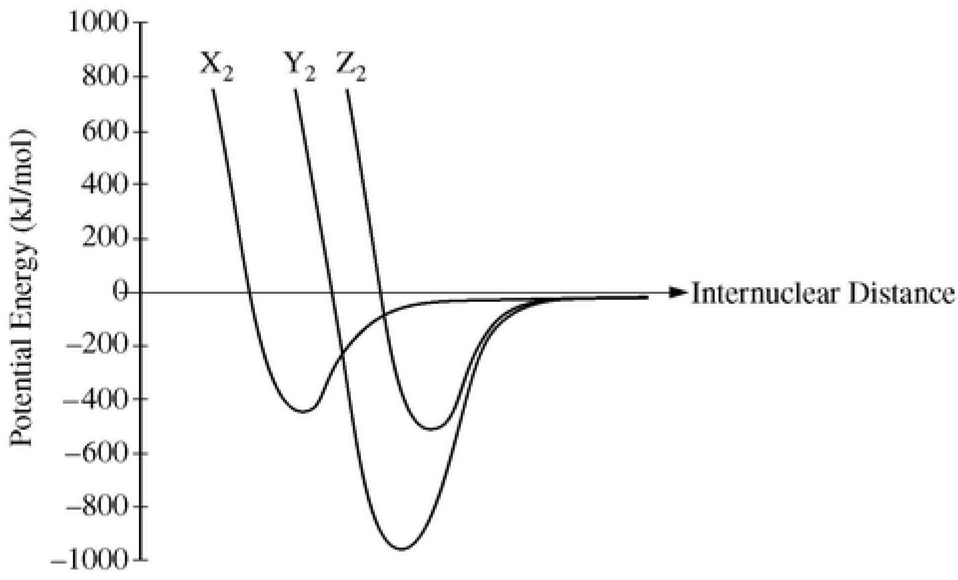
9.



The potential energy as a function of internuclear distance for three diatomic molecules, X_2, Y_2, and Z_2, is shown in the graph above. Based on the data in the graph, which of the following correctly identifies the diatomic molecules, X_2, Y_2, and Z_2?

(A)

X_2	Y_2	Z_2
H_2	N_2	O_2

(B)

X_2	Y_2	Z_2
H_2	O_2	N_2

(C)

X_2	Y_2	Z_2
N_2	O_2	H_2

(D)

X_2	Y_2	Z_2
O_2	H_2	N_2

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10. Which of the following substances has bonds with the greatest ionic character?

(A) CCl_4

(B) $CuCl_2$

(C) $CaCl_2$

(D) NCl_3

11.

$$PCl_3, Cl_2, CaCl_2$$

Which of the following correctly ranks the substances listed above in order of <u>increasing</u> ionic character in their bonding?

(A) Cl_2, PCl_3, $CaCl_2$

(B) Cl_2, $CaCl_2$, PCl_3

(C) $CaCl_2$, PCl_3, Cl_2

(D) $CaCl_2$, Cl_2, PCl_3

12. The lattice energy of a salt is related to the energy required to separate the ions. For which of the following pairs of ions is the energy that is required to separate the ions largest? (Assume that the distance between the ions in each pair is equal to the sum of the ionic radii.)

(A) $Na^+(g)$ and $Cl^-(g)$

(B) $Cs^+(g)$ and $Br^-(g)$

(C) $Mg^{2+}(g)$ and $O^{2-}(g)$

(D) $Ca^{2+}(g)$ and $O^{2-}(g)$

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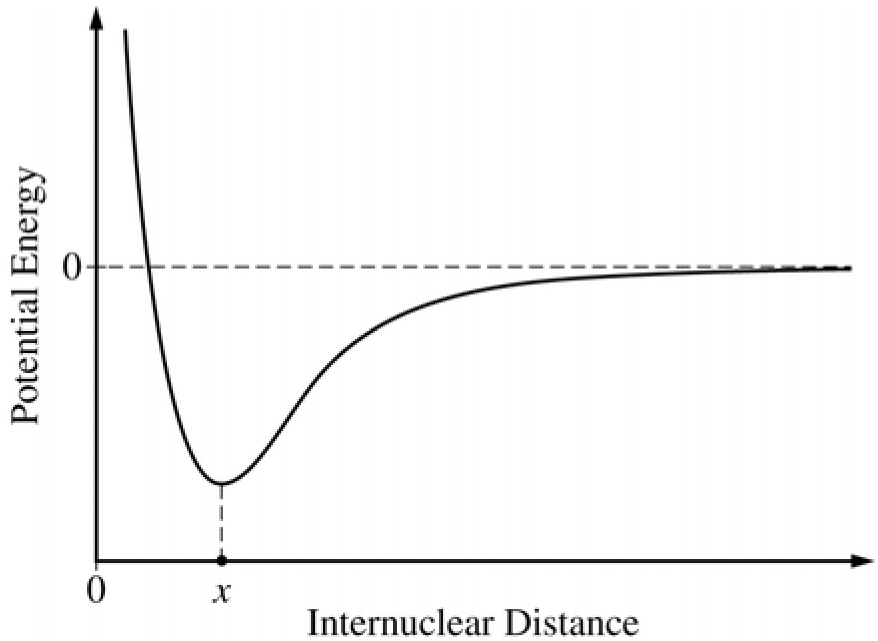
13.



The potential energy of a system of two atoms as a function of their internuclear distance is shown in the diagram above. Which of the following is true regarding the forces between the atoms when their internuclear distance is x?

(A) The attractive and repulsive forces are balanced, so the atoms will maintain an average internuclear distance x.

(B) There is a net repulsive force pushing the atoms apart, so the atoms will move further apart.

(C) There is a net attractive force pulling the atoms together, so the atoms will move closer together.

(D) It cannot be determined whether the forces between atoms are balanced, attractive, or repulsive, because the diagram shows only the potential energy

14.

Molecule	Bond Dissociation Energy (kJ / mol)
Br_2	193
I_2	151

The table above shows the bond dissociation energies for Br_2 and I_2 at 298 K. Which of the following correctly shows the potential energy as a function of internuclear distance for two bromine atoms and two iodine atoms?

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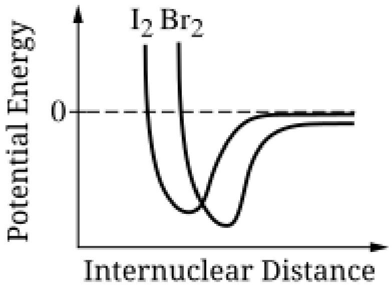
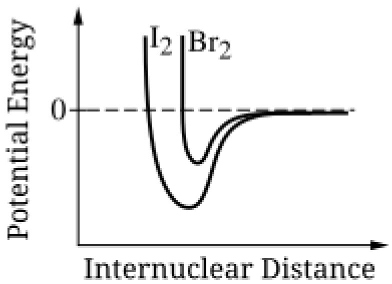
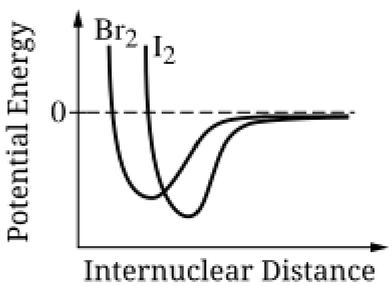
(A)



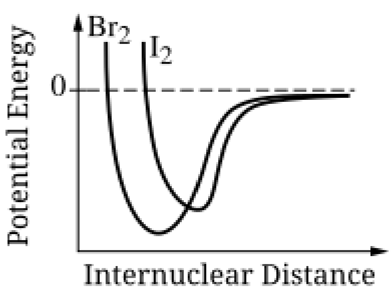
(B)



(C)



(D)



15. Of the following compounds, which is the most ionic?

(A) $SiCl_4$

(B) BrCl

(C) PCl_3

(D) Cl_2O

(E) $CaCl_2$

16. Of the following single bonds, which is the LEAST polar?

(A) N—H
(B) H—F
(C) O—F
(D) I—F
(E) O—H

17. The melting point of MgO is higher than that of NaF. Explanations for this observation include which of the following?

I. Mg^{2+} is more positively charged than Na^{+}.

II. O^{2-} is more negatively charged than F^{-}.

III. The O^{2-} ion is smaller than the F^{-} ion.

(A) II only
(B) I and II only
(C) I and III only
(D) II and III only
(E) I, II, and III